

Gildan Data Presentation

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Problem Framing

Our challenge was to produce retail sales (POS) forecasts that are consistent across hierarchical levels:

1. The Supply planning team requires top-line accuracy at the style level for materials and capacity decisions
2. The Operations team, which includes cutting and sewing, requires size-level SKU forecasts for their production planning

Our ultimate goal was to generate the optimal SKU-level POS forecast for January 2025 through December 2025 using hierarchical reconciliation to ensure consistency across the style, product group, and total portfolio levels.

Data Overview

Structural Findings

- Strong seasonality
- Higher variance at SKU level → smoother aggregate

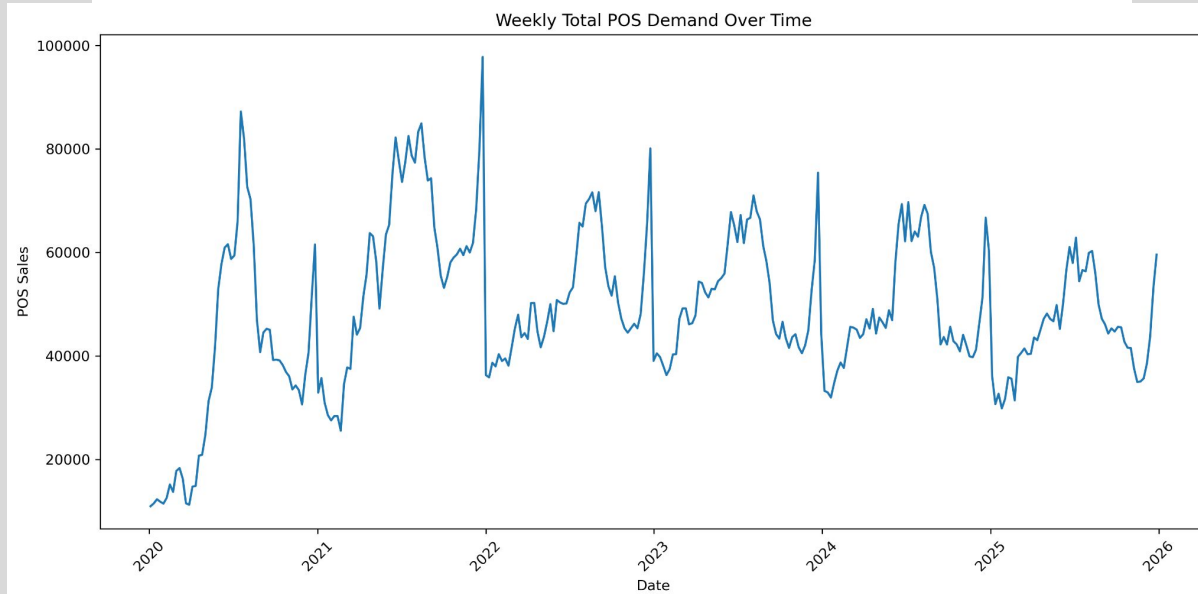
Variables Included

- Price, negative correlation with demand
- Shipments, upstream supply signal
- RetailerInventory, on hand stock
- Calendar features, week-of-year, lag features

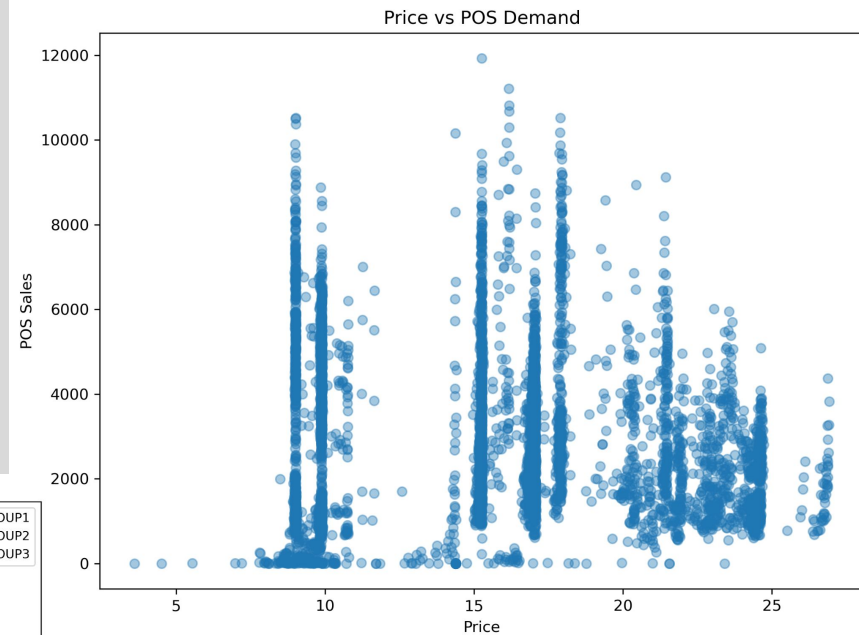
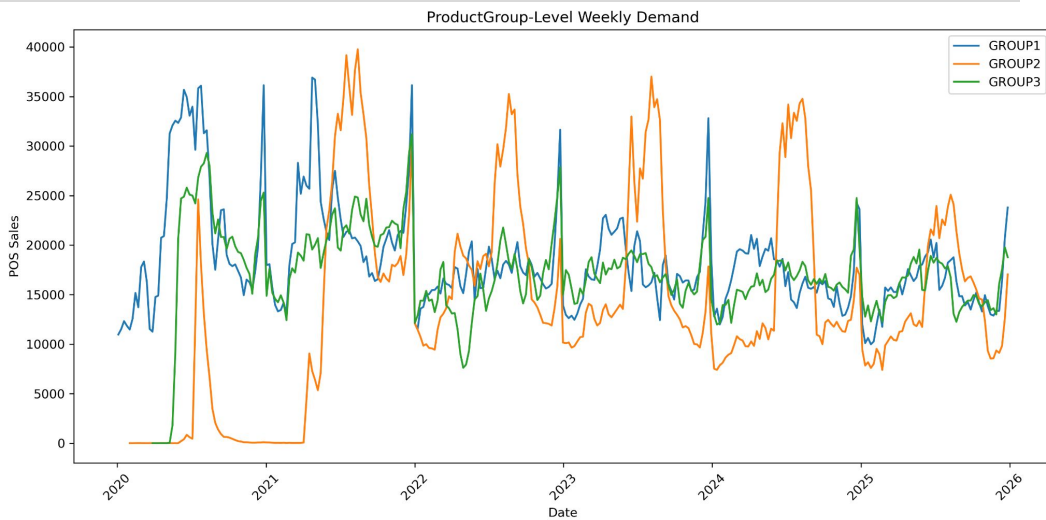
Note: All 2025 data was held out of training and used as the test data set for evaluation.

Descriptive Statistics Overview

	Description	Mean	Std Dev	Min	Max
POS	Weekly point-of-sale demand	2854.61	1933.90	-1.0	11929.00
Price	Retail selling price	15.49	5.11	3.6	26.93
Shipments	Units shipped to retailers	4084.50	4894.82	0.0	63605.00
RetailerInventory	Retail inventory levels	13327.18	7255.34	-1.0	56699.00
PtsofDistrib	Distribution / door coverage	2232.97	500.27	0.0	7067.00



Data Overview



Modeling Approach

[BASELINE MODELS](#)

Naive - Uses last week's sales as the next prediction. RMSE around 1,800. Serves as performance baseline.

ARIMA - Started modeling at a SKU level (bottom-up approach), captured short term correlation but struggled with size-level noise.

[ADVANCED MODELS](#)

Random Forest - First machine learning model, ran a grid search for optimal hyperparameter tuning (TimeSeriesSplit).

XGBoost - Two-stage grid search, better with nonlinear relationships.

GRU - Deep learning model, sequence model, 12-week lookback per SKU, dropout regularization

Modeling Approach

FEATURE IMPORTANCE

Lag_1 - prior week's sales consistently #1 predictor across ML models

Price - increases in price consistently impact demand in a negative way

RetailorInventory - higher on-hand inventory correlated with greater in-stock lift and sales

Week-of-Year - mid year lags naturally captured back-to-school and holiday lift windows

ACCURACY-INTERPRETABILITY TRADEOFF

Machine learning models deliver significantly better results than baseline → at the cost of model explainability.

-Can be mitigated by using SHAP.

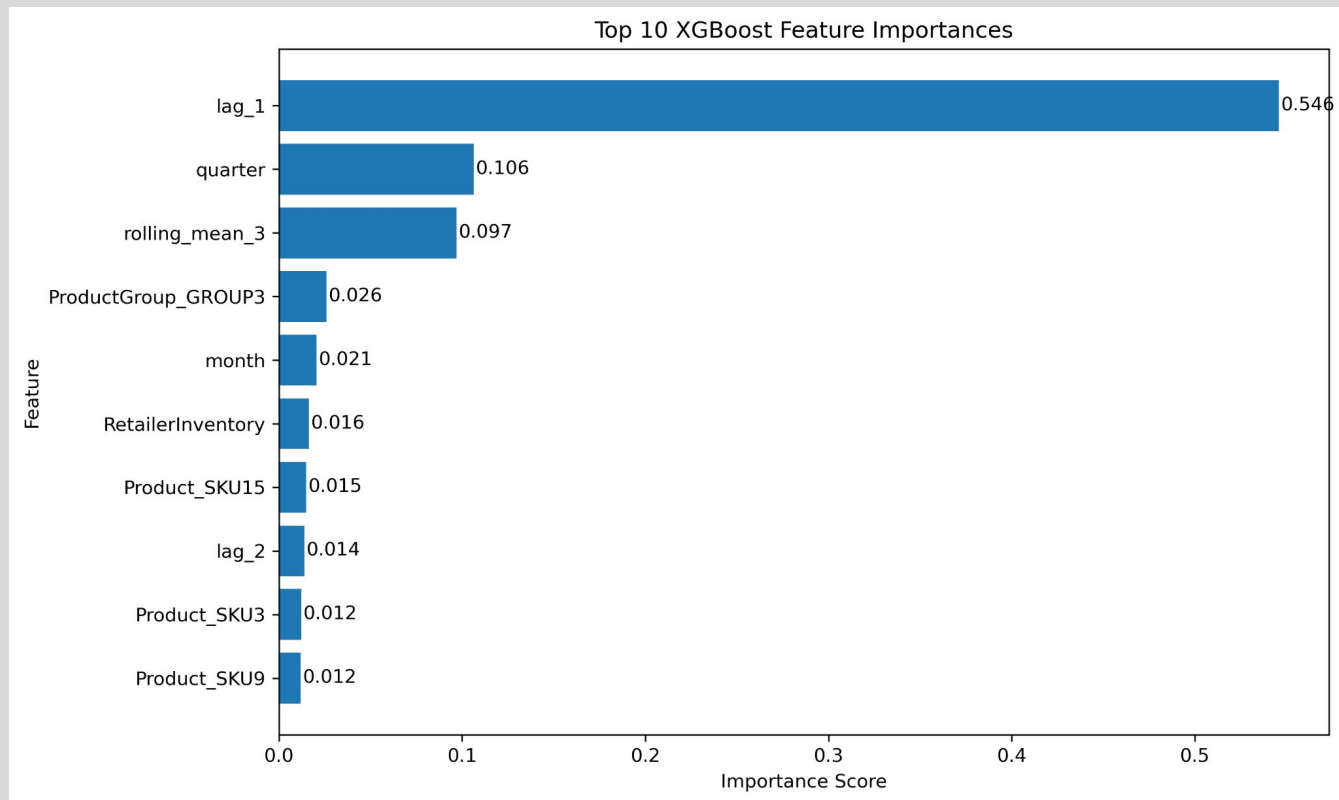
GRU offered sequence learning but showed recursive drift. Better for short-term and not for supply planning.

Model Selected for Analysis: XGBoost

-High accuracy and interpretation via SHAP.

XGBoost Feature Importance

- **lag_1 (0.546)** — Previous week demand was by far the strongest predictor, indicating that short-term sales momentum and demand persistence heavily influenced future POS sales.
- **quarter (0.106)** — Quarterly seasonality played a meaningful role in forecasting accuracy, suggesting that demand patterns changed systematically across different times of the year.
- **rolling_mean_3 (0.097)** — The recent 3-week average demand helped stabilize forecasts and capture short-term trend direction beyond a single lag observation.



Hierarchical Reconciliation

Bottom-Up Reconciliation

- Forecasted directly at the SKU level, then aggregated upward
- Preserved SKU-level demand variability and operational detail
- Produced the strongest overall forecasting performance

Top-Down Reconciliation (XGBoost Only)

- Forecasted at the Total Portfolio level, then allocated downward to SKUs
- Maintained identical portfolio totals as Bottom-Up
- Reduced SKU-level specificity and lower-level accuracy

Evaluation Levels

- SKU Level
- ProductGroup Level
- Total Portfolio Level

Evaluation Metrics

- WMAPE
- MAPE
- RMSE
- Bias
- R^2

Hierarchical Forecast Performance Evaluation

SKU-Level Evaluation Results

	Model	WMAPE	MAPE	RMSE	Bias	R ²
0	Random Forest	35.67	47.51	1101.94	29.12	0.464
1	GRU	36.82	53.74	1195.39	24.19	0.370
2	XGBoost Bottom-Up	19.31	25.69	681.78	5.46	0.795
3	XGBoost Top-Down	21.58	36.32	720.90	5.46	0.771

SKU Level: Bottom-Up XGBoost achieved the best operational accuracy - lowest WMAPE (19.31%) and highest R² (0.795)

ProductGroup Level: Bottom-Up XGBoost also performed best at the aggregated group level, preserving SKU-level demand structure

Total Portfolio Level: Top-Down and Bottom-Up XGBoost produced identical portfolio totals, showing reconciliation mainly impacts lower-level forecast allocation

ProductGroup-Level Evaluation Results

	Model	WMAPE	MAPE	RMSE	Bias	R ²
0	Random Forest	33.42	39.66	5872.22	29.12	-1.530
1	GRU	34.13	39.68	6098.86	24.19	-1.729
2	XGBoost Bottom-Up	13.40	14.75	2942.72	5.46	0.365
3	XGBoost Top-Down	18.84	19.49	3572.56	5.46	0.064

Total Portfolio Evaluation Results

	Model	WMAPE	MAPE	RMSE	Bias	R ²
0	Random Forest	30.07	34.43	15739.21	29.12	-2.298
1	GRU	26.90	31.25	14333.59	24.19	-1.735
2	XGBoost Bottom-Up	10.91	11.68	6742.43	5.46	0.395
3	XGBoost Top-Down	10.91	11.68	6742.43	5.46	0.395

Key Findings

- XGBoost outperformed RF and GRU across all hierarchy levels.
- Bottom-Up reconciliation preserved SKU-level demand better than Top-Down.
- Forecast accuracy improved at higher aggregation levels.
- Forecast accuracy declined as the forecasting horizon became longer.

Operational Insight:

Bottom-Up forecasting is better suited for inventory and production planning because it preserves SKU-level demand behavior while still producing stable, aggregated forecasts.

XGBoost Results - Bottom Up

SKU-Level XGBoost Performance

	SKU	WMAPE	MAPE	RMSE	Bias	R ²
13	SKU14	5.93	6.08	327.59	0.79	0.759
14	SKU15	7.47	7.25	536.22	-5.86	0.568
15	SKU16	8.38	8.58	336.65	1.60	0.493
12	SKU13	9.58	9.66	93.01	-0.08	0.396
4	SKU5	14.56	15.03	311.97	0.11	0.255
2	SKU3	14.99	16.20	698.43	10.08	-0.019
6	SKU10	18.89	22.64	728.00	16.06	0.573
5	SKU6	20.42	20.82	455.84	0.43	-0.852
11	SKU9	20.86	20.37	1018.73	10.79	0.415
3	SKU4	21.78	23.27	889.16	20.36	-1.200
1	SKU2	27.82	29.49	1211.26	8.10	-1.939
10	SKU8	28.01	29.81	1047.88	25.74	0.128
7	SKU11	28.68	29.13	551.24	-10.35	-0.152
16	SKU17	28.96	27.93	403.97	-27.41	-3.549
8	SKU12	33.40	35.02	576.54	-16.64	-0.061
0	SKU1	36.05	35.64	708.24	-13.21	-1.215
17	SKU18	47.84	62.55	161.63	35.46	-1.238
9	SKU7	52.16	62.95	897.92	52.06	-0.878

ProductGroup-Level XGBoost Performance

	ProductGroup	WMAPE	MAPE	RMSE	Bias	R ²
2	GROUP3	5.81	5.57	1136.75	-3.10	0.716
1	GROUP2	16.67	19.96	3414.75	13.32	0.564
0	GROUP1	18.04	18.72	3609.17	6.90	-0.582

Group 3 achieved the strongest group-level performance

- WMAPE: 5.81%
- R²: 0.716

More volatile SKUs remained difficult to forecast:

- SKUs 14–16 produced the strongest SKU-level accuracy
 - WMAPE between 5–9%
- Lower-volume SKUs such as SKU7 and 18 showed significantly higher error rates, highlighting variability at the granular demand level.

Bottom-Up reconciliation preserved local demand behavior:

- Forecasting directly at the SKU level captured short-term demand patterns more effectively before aggregation upward to ProductGroups.

XGBoost Results Top-Down

SKU-Level Top-Down XGBoost Performance						
	SKU	WMAPE	MAPE	RMSE	Bias	R ²
5	SKU14	11.52	11.56	663.37	-4.84	0.011
6	SKU15	12.87	13.09	913.85	-4.10	-0.254
8	SKU17	15.18	16.82	260.77	5.38	-0.895
7	SKU16	16.94	17.12	621.47	5.34	-0.726
2	SKU11	20.44	20.07	415.81	-7.77	0.344
14	SKU6	20.47	20.98	466.86	-13.68	-0.942
12	SKU4	20.57	21.69	809.64	18.79	-0.824
1	SKU10	21.31	21.37	849.45	-5.00	0.419
17	SKU9	21.88	23.68	1001.27	0.25	0.435
0	SKU1	22.26	27.45	481.32	14.99	-0.023
10	SKU2	23.88	26.87	948.99	23.03	-0.804
16	SKU8	23.97	25.11	880.97	-2.68	0.384
4	SKU13	25.44	26.47	251.43	21.87	-3.416
13	SKU5	26.62	28.26	577.95	25.91	-1.558
11	SKU3	27.00	28.70	1240.25	25.79	-2.212
15	SKU7	29.51	32.01	538.02	-0.49	0.326
3	SKU12	33.35	30.78	626.40	-28.23	-0.253
9	SKU18	193.46	261.70	573.64	193.46	-27.193

ProductGroup-Level Top-Down XGBoost Performance

	ProductGroup	WMAPE	MAPE	RMSE	Bias	R ²
2	GROUP3	14.72	14.90	2660.57	3.14	-0.554
0	GROUP1	20.00	21.37	3894.53	17.88	-0.842
1	GROUP2	22.10	22.22	4005.44	-5.62	0.400

Top-Down reconciliation reduced SKU-level precision:

- SKU-level WMAPE increased versus Bottom-Up reconciliation, showing weaker lower-level allocation accuracy.

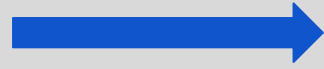
ProductGroup performance weakened after allocation:

- GROUP3 remained the strongest-performing group
- WMAPE: 14.72%

Several SKUs experienced significant forecast distortion:

- SKU18 produced extremely high forecasting error
- WMAPE: 193.46%
- Bias: 193.46%

Modeling



Results

Insights

Demand Lever Analysis

- Price is the most controllable lever: Price reductions drive the largest positive demand response across all SKUs
- Retailer Inventory: ensuring sufficient on-hand stock at retail is the second-strongest positive driver, stock-outs are a hidden demand killer
- Momentum (lag_1) dominates short-term forecasts; seasonal peaks around back-to-school and holiday periods offer predictable demand windows to target

POS to Shipment Forecast Translation

- Shipment forecast can be applied as a layer on top of our SKU-level XGBoost output
- Use quarterly aggregation XGBoost forecast as the input signal: minimizes weekly noise while retaining seasonality and direction
- Shipment lag adjustment: retailers typically replenish 2-4 weeks after POS; shift the forecast forward to align manufacturer shipment timing

Recommendations

Decision Horizon	Forecast Signal	Operational Use
Production (2-3 months)	4-week rolling SKU level XGBoost forecast	Most granular and current signal for the cutting, sewing, and inventory allocations
Capacity (5-7 months)	Quarterly Product Group XGBoost forecast	This is the strongest forecasting level for labor and factory capacity planning
Materials (8-12 months)	Annual style-level aggregated forecast	Due to long-horizon uncertainty, must use forecast ranges and not point estimates

Capacity Constraint Scenarios

90% Capacity: Prioritize high-revenue SKUs with strongest demand confidence; defer low-velocity sizes to protect core assortment

110% Capacity: Pre-build stable core sizes with low demand variance; avoid overproduction of niche SKUs with high SLOB risk

XGBoost exhibits a **+5.46% under-forecast bias** → adjust production plans upward by ~5% to improve service levels and reduce stockout risk

Additional Data to Improve Forecasting:

1. Syndicated Retail POS Data
2. Retailer Promotional Calendars
3. Consumer Sentiment and Search Trends
4. Macroeconomic Indicators
5. Door Count and Distribution Velocity

Expanding the model beyond internal data would significantly strengthen forecast quality by improving our ability to distinguish supply-side constraints from true underlying consumer demand. Incorporating additional external signals would help reduce forecast bias and limit the recursive error accumulation that becomes more severe at longer horizons. These enhancements would be especially valuable for quarterly and annual forecasts, where inaccurate planning decisions create the greatest operational and financial impact.

New Demand Sensing Architecture

Since the XGBoost model performs strongly at the quarterly horizon, this is a reliable signal for capacity and materials planning

Adding a near-term demand sensing layer on top of the existing XGBoost framework will help create a faster response to short-term demand shifts.

This model would utilize daily and weekly retailer POS feeds, inventory updates and short-term trend signals to be one steps ahead and reduce production volatility and last-minute order changes.

Thank You!